Fuel Control Unit – Maintenance Practices

The Fuel Control Unit (FCU), as received from the OEM or overhaul facility, must be reconfigured to function correctly on the Twin Otter, Series 400.

1. Removal/Installation

CAUTION ANY DAMAGED OR UNACCEPTABLE HARDWARE SHOULD BE REPLACED PRIOR TO INSTALLATION ON NEW FCU.

A. To remove power plant, refer to 71-00-00, Power Plant (Mod 6/2012) – General Data.

B. To remove existing FCU, refer to P&WC PT6A–34 Engine Maintenance Manual, Chapter 73-20-00.

C. Remove from existing FCU the following, refer to Figure 1.

1) Remove the FCU Arm, washer and nut off of the FCU Spindle Shaft. Retain for installation on new FCU.

2) Remove the screws (2), washers (2) and nuts (2) from the bottom end of the FCU Shutoff Control Levers.

3) Remove the cotter pins from the FCU Intermediate (shaft) Pin, and washers from the outside of the FCU Shutoff Control Levers. Retain hardware for installation on new FCU.

4) Remove the nut, washers (2), bolt and spacer that keeps the FCU Shutoff Control Levers aligned. Retain hardware for installation on new FCU.

5) Remove the FCU Shutoff Control Levers from the right and left hand sides of the FCU Intermediate Lever. Retain FCU Shutoff Control Levers, FCU Intermediate (shaft) Pin and associated hardware for reinstallation on new FCU.

D. Rework new FCU as follows:

1) Cut off safety cable, support Pratt & Whitney supplied FCU arm and remove the Pratt & Whitney supplied extension nut. Remove Pratt & Whitney supplied FCU arm and retain with the Pratt & Whitney supplied extension nut to be reinstalled on old FCU.

CAUTION DO NOT REMOVE SERRATED SPACER FROM THE FCU SPINDLE SHAFT.

2) Temporarily install FCU arm, washer and nut removed from the old FCU, refer to Para 1.C.1).

3) Remove the Pratt & Whitney supplied stop bolt from new FCU. Retain to be reinstalled on old FCU.

4) Remove the cotter pins (2) from the FCU Intermediate Lever (shaft) Pin.

5) Remove the Pratt & Whitney supplied nut and spacer from High Idle Rod. Retain to be reinstalled on old FCU.

6) Carefully withdraw the FCU Intermediate (shaft) Pin until R/H side of FCU Intermediate Lever can be slid forward to free the High Idle Rod from the Intermediate Lever.

7) Reposition FCU Intermediate Lever and reinsert the FCU Intermediate Lever (shaft) Pin. Remove cotter pin from base of High Idle Rod, remove High Idle Rod. Retain to be reinstalled on old FCU.

8) Hold FCU Intermediate Lever in position and remove the FCU Intermediate Lever (shaft) Pin. Install the FCU Shutoff Control Lever on left side of Intermediate Lever with flat side to the rear and reinsert the FCU Intermediate Lever (shaft) Pin. Install other FCU Shutoff Control Lever on right side of Intermediate Lever with flat side facing forward, refer to Figure 1.

9) Using a 3/16” bolt or drill bit to help align the two FCU Shutoff Control Levers, install spacer, bolt, washer and nut.

10) Install screws (2), washers (2), nuts (2) in the bottom hole of each FCU Shutoff Control Lever.

Note The flat side of the FCU Shutoff Control Levers are facing forward.

11) Install cotter pins (2) in FCU Intermediate Lever (shaft) Pin.

12) Remove alignment bolt/bit drill.

13) Install screw and nut for new stop bolt.

14) Attach rigging gauge SD12556 using a temporary washer in between gauge and lever, to maintain alignment. Adjust FCU arm so that rig pin on gauge slides freely through the hole on the lever.

Note There should be no friction on the rig pin.

15) Readjust serrated spacer as required to eliminate any friction. Once adjusted, mark an index line across FCU shaft, serrated spacer and lever for ground run rigging reference.

16) Tighten nut and install MS24665–138 cotter pin (may require removal for ground runs).

17) Hold Intermediate Lever in closed position and adjust stop bolt until light contact with lever occurs. Adjust “null” dimension between stop bolt and throttle pick-up position so that 0.020” gap is achieved for static rigging ONLY.

18) Tighten jam nut.

E. Reinstall retained parts on the old FCU as follows:

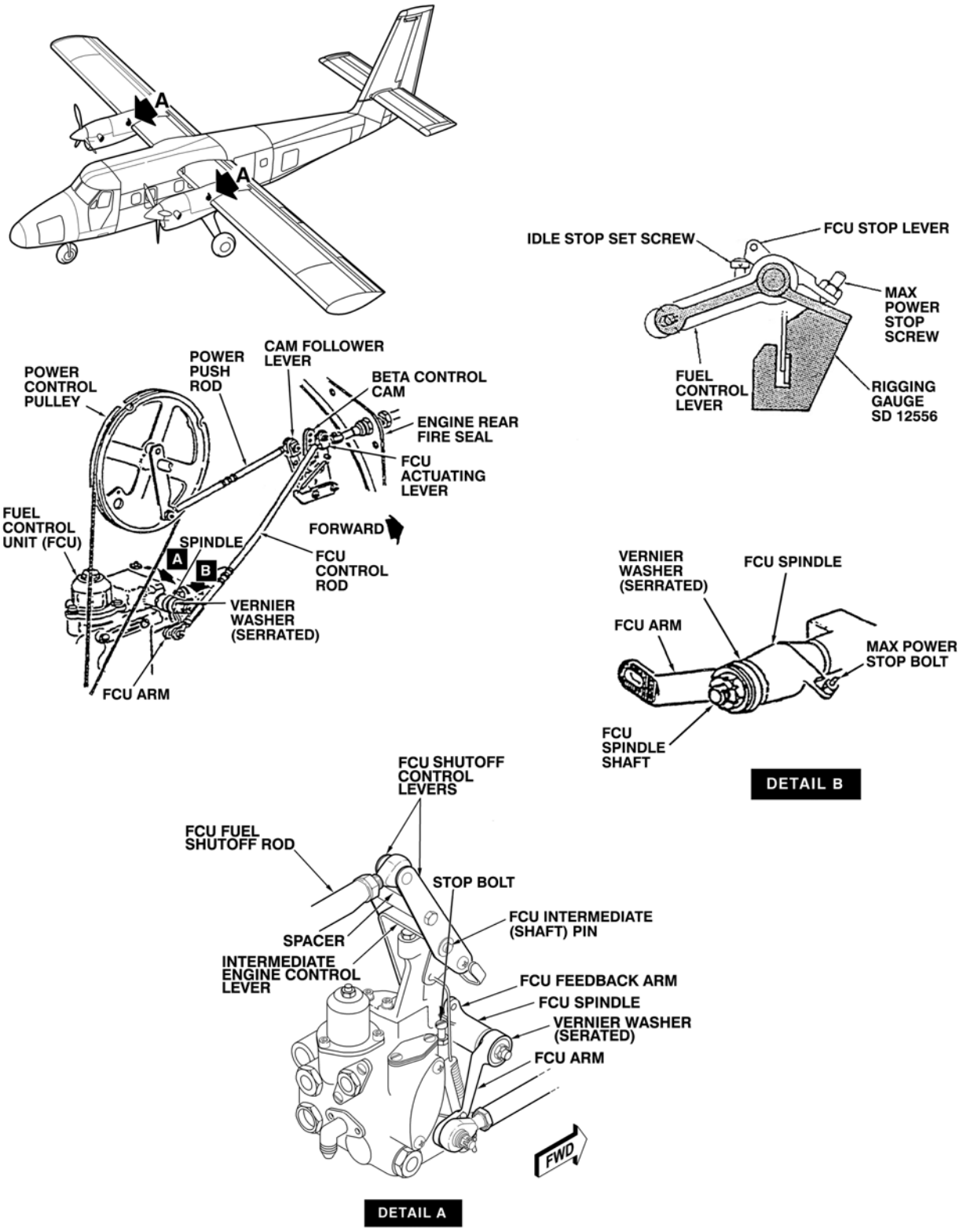
1) Install the Pratt & Whitney supplied FCU arm and supplied extension nut on old FCU.

2) Install the Pratt & Whitney supplied stop bolt on old FCU.

3) Install the High Idle Rod and cotter pins (2) on old FCU.

F. To install new FCU, refer to P&WC PT6A–34 Engine Maintenance Manual, Chapter 73-20-00.

G. To install power plant, refer to 71-00-00, Power Plant (Mod 6/2012) – General Data.



Power Lever Control Rigging
Figure 1

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